Assignment 1 Report

LinearRegression().fit()

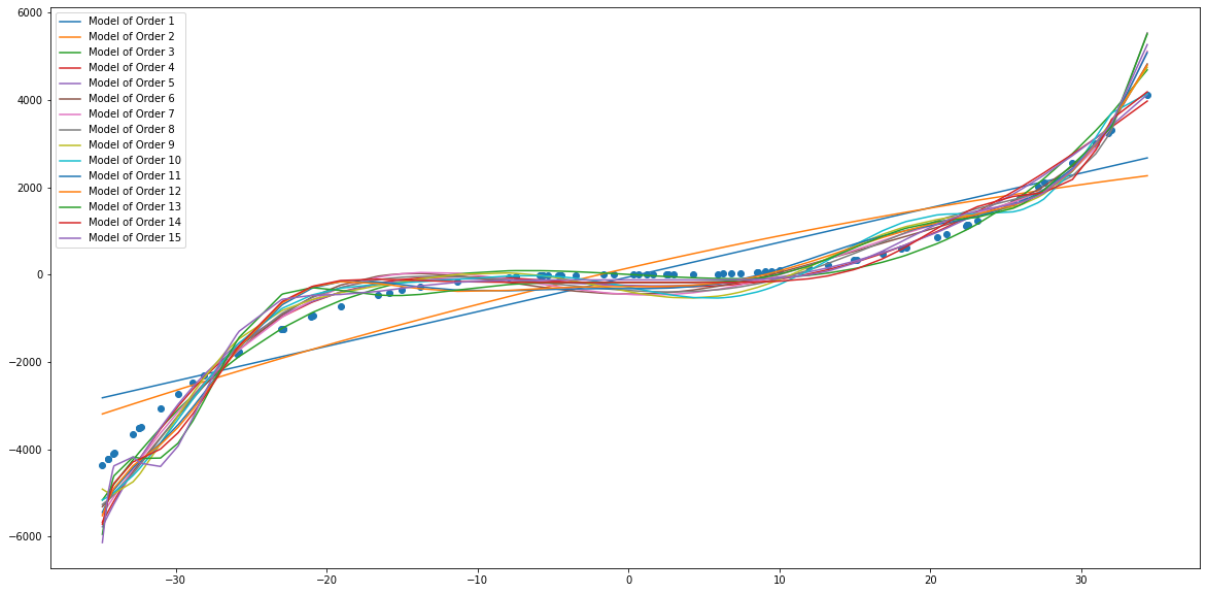
A linear regression model represents a linear equation which has a certain set of inputs (x) and predicted outputs for those inputs (y).

$y = B_0 + B_1(x)$

Here, B_0 is the bias coefficient/intercept.

The intercept is the expected mean value of Y when all X equals to 0.

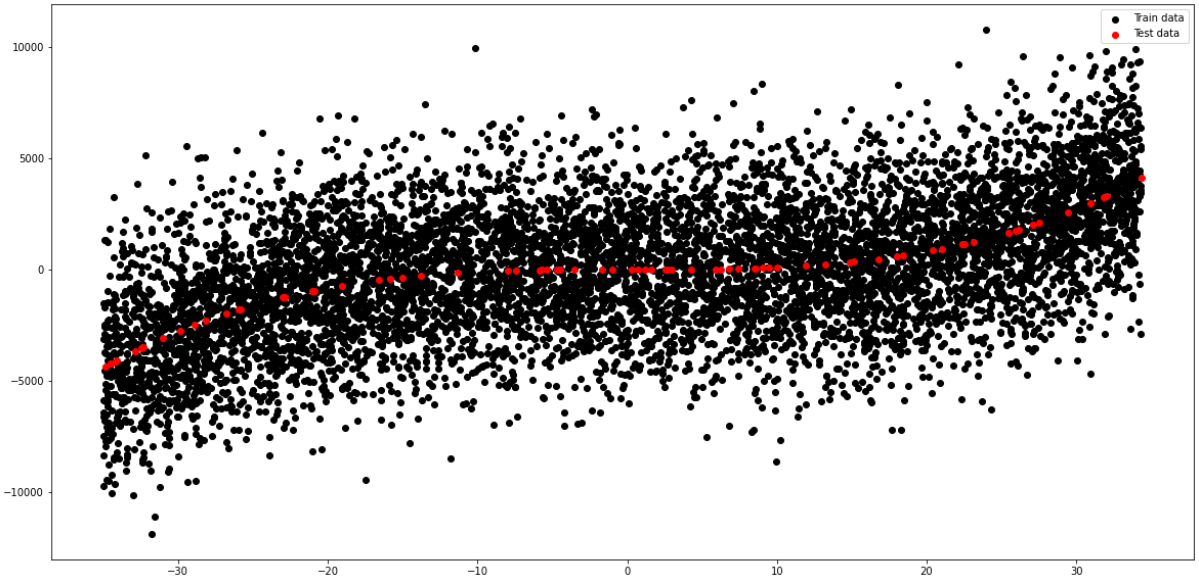
.fit() function finds the optimal values of the intercepts where the arguments are existing inputs and outputs. An estimator is fit onto the model to predict the outputs of the unseen inputs. A classifier is an estimator instance which is fitted to the model and its purpose is to learn from the model. The .fit() function fits any instance of the linear regression.



Irreducible Errors

The irreducible error values hardly change. They are very close to 0.

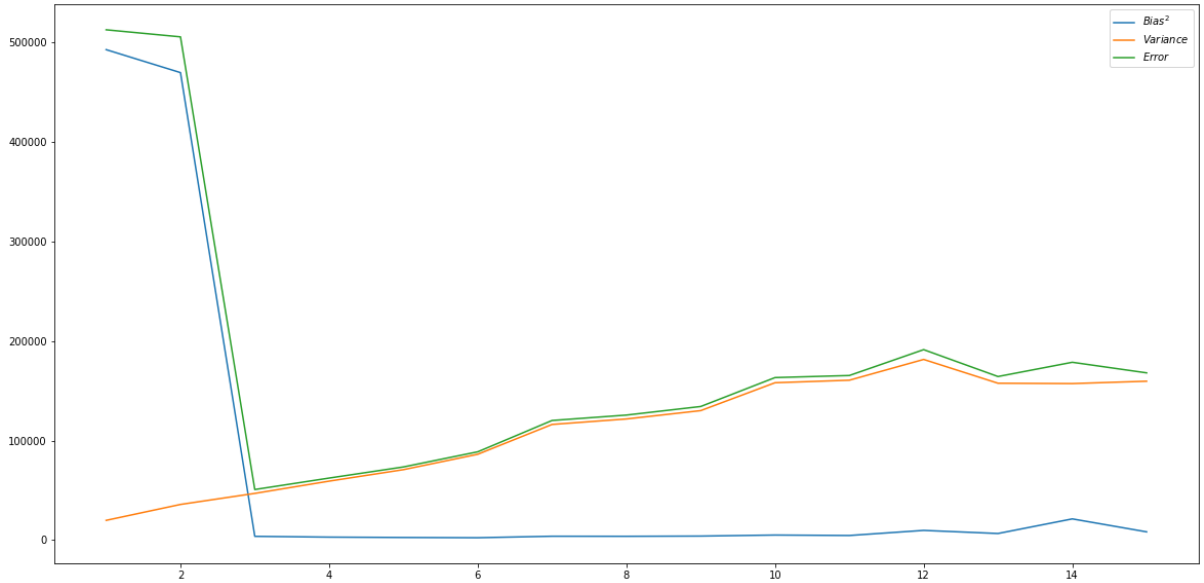
The irreducible error does not change, as it is a property of the test data. We assume that noise is normally distributed, and our irreducible error $E_{irreducible} = \sigma^2$. This value will be consistent across realisations of the testing data, and across polynomial orders. Looking at the scatterplot below, we can see that the test data (in red) seems to exactly fit a polynomial, which leads us to believe that there was minimal noise introduced into the test data. This explains why the observed irreducible error is constant, and very close to 0.



	Irreducible error
1	-9.822543e-11
2	5.820766e-11
3	0.000000e+00
4	7.275958e-12
5	-1.455192e-11
6	1.455192e-11
7	2.910383e-11
8	2.910383e-11
9	1.455192e-11
10	-2.910383e-11
11	-5.820766e-11
12	0.000000e+00
13	0.000000e+00
14	2.910383e-11
15	2.910383e-11

Bias vs Variance

As the order changes, squared bias decreases and variance increases.



From the above graph, we can conclude the following:

High Bias, Low Variance -

When a model (polynomial order) has high bias and low variance, it 'underfits' the data. We can see this as models with low orders have high biases. This is because they tend to simplify the estimator that are very similar to each other and result in underfitting.

High Variance, Low Bias -

When a model (polynomial order) has high variance and low bias, it 'overfits' the data. We can see this as models with high orders have high variances. This is because they tend to create complex estimators that vary a lot from each other and result in overfitting.

Ideal model -

As we can see, the lowest error (mean squared) is observed at order 3. This is not only for this particular run of the training data split, order 3 gives the best results consistently. However orders from 3-6 have acceptable errors, and could all potentially be used to estimate the function.

Dataset -

The provided test set lacks noise, and hence has irreducible error very close to 0. We can also reasonably conclude that the data given is of order three with respect to the input parameter, as the lowest error is obtained when our order is three, and the shape of the graph also indicates that $y = ax^3 + bx^2 + cx + d$.